

# Advanced Quantum Theory: Notes

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# 1 LC Chain: Harmonic Oscillators and Waves

An **LC chain** (or LC ladder network) is a classic physical system used to represent a series of coupled **harmonic oscillators**. It models how energy—like sound or heat—moves through a solid material, treating inductors ( $L$ ) and capacitors ( $C$ ) as electrical versions of masses and springs.

## The Mechanical Analogy

| Electrical (LC Chain) | Mechanical Equivalent | Physical Role                                   |
|-----------------------|-----------------------|-------------------------------------------------|
| Inductor ( $L$ )      | Mass ( $m$ )          | Represents inertia; resists changes in current/ |
| Capacitor ( $C$ )     | Spring ( $1/k$ )      | Represents elasticity; stores potential energy  |
| Current ( $I$ )       | Velocity ( $v$ )      | The rate of flow/movement                       |
| Voltage ( $V$ )       | Force ( $F$ )         | The “push” that causes movement                 |

## The Governing Equation

For a single unit in the chain, the relationship between the charge ( $q$ ) and the components follows the standard harmonic oscillator equation:

$$L \frac{d^2 q}{dt^2} + \frac{1}{C} q = 0 \quad (1)$$

When linked together, the change in one “cell” affects the next, creating collective motion described by a system of coupled differential equations.

## Expressions for $\phi_k$ and $\pi_k$

In a finite system of length  $L$ , fields are represented as sums over discrete wavevectors  $k$ . The Fourier coefficients are defined as:

$$\phi_k = \frac{1}{\sqrt{L}} \int dx e^{-ikx} \phi(x) \quad (2)$$

$$\pi_k = \frac{1}{\sqrt{L}} \int dx e^{ikx} \pi(x) \quad (3)$$

Using the standard commutation relation  $[\phi(x), \pi(y)] = i\delta(x - y)$ , these transformed operators satisfy:

- $\phi_k^\dagger = \phi_{-k}$
- $\pi_k^\dagger = \pi_{-k}$
- $[\phi_k, \pi_{k'}] = i\delta_{k,k'}$

## The Hamiltonian

The Hamiltonian for a chain of harmonic oscillators (or a free scalar field) in real space is:

$$H = \int dx \left[ \frac{1}{2} \pi^2(x) + \frac{1}{2} (\nabla \phi(x))^2 + \frac{1}{2} m^2 \phi^2(x) \right] \quad (4)$$

By substituting the Fourier expansions  $\phi(x) = \frac{1}{\sqrt{L}} \sum_k e^{ikx} \phi_k$  and  $\pi(x) = \frac{1}{\sqrt{L}} \sum_k e^{-ikx} \pi_k$ , the Hamiltonian diagonalizes into a sum of independent oscillators:

$$H = \sum_k \left( \frac{1}{2} \pi_k \pi_{-k} + \frac{1}{2} \omega_k^2 \phi_k \phi_{-k} \right) \quad (5)$$

where the dispersion relation  $\omega_k$  is determined by the physical properties of the chain (like the  $L$  and  $C$  values).

## Creation and Annihilation Operators

We define  $a_k$  and  $a_k^\dagger$  in terms of the Fourier operators:

$$a_k = \sqrt{\frac{\omega_k}{2}} \left( \phi_k + \frac{i}{\omega_k} \pi_{-k} \right) \quad (6)$$

$$a_k^\dagger = \sqrt{\frac{\omega_k}{2}} \left( \phi_{-k} - \frac{i}{\omega_k} \pi_k \right) \quad (7)$$

The commutator  $[a_k, a_{k'}^\dagger]$  expands as:

$$[a_k, a_{k'}^\dagger] = \frac{\sqrt{\omega_k \omega_{k'}}}{2} \left[ \phi_k + \frac{i}{\omega_k} \pi_{-k}, \phi_{-k'} - \frac{i}{\omega_{k'}} \pi_{k'} \right] \quad (8)$$

Expanding the brackets using  $[\phi_k, \phi_{-k'}] = 0$ ,  $[\pi_{-k}, \pi_{k'}] = 0$ ,  $[\phi_k, -\frac{i}{\omega_{k'}} \pi_{k'}] = \frac{1}{\omega_{k'}} \delta_{k,k'}$ , and  $[\frac{i}{\omega_k} \pi_{-k}, \phi_{-k'}] = \frac{1}{\omega_k} \delta_{k,k'}$ , one obtains:

$$\boxed{[a_k, a_{k'}^\dagger] = \delta_{k,k'}} \quad (9)$$

## 2 Rabi Oscillations

**Rabi oscillations** are the characteristic oscillatory dynamics of a two-level quantum system (like a spin- $\frac{1}{2}$  particle, or an atom with ground and excited states) when it interacts with a coherent oscillatory driving field (such as a laser or an oscillating magnetic field).

### Core Idea

Consider a two-level system with states  $|0\rangle$  (ground) and  $|1\rangle$  (excited). When driven by a near-resonant oscillatory field, the system **coherently oscillates** between them. The probability of finding the system in the excited state varies sinusoidally in time.

## Mathematical Form

The excited-state probability is:

$$P_1(t) = \frac{\lambda^2}{\delta^2 + \lambda^2} \sin^2\left(\frac{\Omega_R t}{2}\right) \quad (10)$$

where:

- $\lambda$  = coupling strength (interaction with the field),
- $\delta = \omega - \Delta$  = detuning (difference between drive frequency and transition frequency),
- $\Omega_R = \sqrt{\delta^2 + \lambda^2}$  = **Rabi frequency**.

## Behavior

- **At resonance** ( $\delta = 0$ ): The system oscillates fully between ground and excited states:

$$P_1(t) = \sin^2\left(\frac{\lambda}{2}t\right) \quad (11)$$

- **Off resonance** ( $\delta \neq 0$ ): Oscillations still occur, but with reduced amplitude and faster frequency. The system never fully reaches the excited state.

## Derivation via Rotating Frame

Starting from the Hamiltonian (after shifting the zero of energy):

$$H = \begin{pmatrix} 0 & \lambda \cos(\omega t) \\ \lambda \cos(\omega t) & -\Delta \end{pmatrix} \quad (12)$$

with the wavefunction  $|\psi(t)\rangle = \begin{pmatrix} c_0(t) \\ c_1(t) \end{pmatrix}$ , the Schrödinger equation gives:

$$i\dot{c}_0 = \lambda \cos(\omega t) c_1, \quad i\dot{c}_1 = \lambda \cos(\omega t) c_0 - \Delta c_1 \quad (13)$$

Define the rotating frame variable  $\tilde{c}_1(t) = c_1(t)e^{i\Delta t}$ . Applying the **Rotating Wave Approximation (RWA)** (dropping fast oscillating terms):

$$\dot{c}_0 = i\frac{\lambda}{2}e^{i(\omega-\Delta)t}\tilde{c}_1, \quad \dot{\tilde{c}}_1 = i\frac{\lambda}{2}e^{-i(\omega-\Delta)t}c_0 \quad (14)$$

Differentiating the second equation yields the second-order equation:

$$\ddot{\tilde{c}}_1 + i\delta\dot{\tilde{c}}_1 + \frac{\lambda^2}{4}\tilde{c}_1 = 0, \quad \delta = \omega - \Delta \quad (15)$$

This is a harmonic oscillator-like equation. The solution gives the **excited state probability**:

$$|c_1(t)|^2 = \frac{\lambda^2}{4\Omega_R^2} \sin^2(\Omega_R t) \quad (16)$$

with the **Rabi frequency**:

$$\Omega_R = \frac{1}{2}\sqrt{\delta^2 + \lambda^2} \quad (17)$$

### 3 The Rotating Wave Approximation (RWA)

The **Rotating Wave Approximation (RWA)** is a simplification technique used in quantum mechanics, especially in quantum optics and magnetic resonance, to make time-dependent Hamiltonians analytically manageable.

#### Core Idea

When a two-level system interacts with an oscillating field, the Hamiltonian contains terms oscillating at different frequencies:

- Some terms oscillate **slowly** (near resonance, frequency difference small).
- Other terms oscillate **very rapidly** (frequency sum large).

**RWA neglects the rapidly oscillating terms**, because their average effect over time is negligible. Only the near-resonant terms are kept.

#### When RWA is Valid

- The driving field frequency  $\omega$  is close to the system's transition frequency  $\Delta$  (near resonance).
- The coupling strength  $\lambda$  is relatively small compared to  $\omega$ .
- Under these conditions, the fast oscillating terms contribute negligibly to the dynamics, while the near-resonant terms dominate.

#### Intuition

Think of RWA as “tuning in” to the beat frequency between the drive and the atom, while ignoring the high-frequency “static” that doesn’t affect long-term dynamics.

### 4 Jaynes–Cummings Model

#### The JC Hamiltonian

$$H = -\Delta\sigma_z + \omega_c a^\dagger a + g(\sigma_+ a + \sigma_- a^\dagger) \quad (18)$$

where:

- $\Delta$ : atomic transition frequency (detuning term),
- $\omega_c$ : cavity photon frequency,
- $g$ : atom–cavity coupling strength,
- Basis:  $|n_{\text{atom}}, n_{\text{photon}}\rangle$  with  $n_{\text{atom}} = 0, 1$  and  $n_{\text{photon}} = 0, 1, 2, \dots$

## Relevant Subspace

Starting with the initial state  $|\psi(0)\rangle = |1, 0\rangle$ , the dynamics are restricted to the **two-dimensional subspace** spanned by:

$$|1, 0\rangle \text{ (excited atom, no photon),} \quad |0, 1\rangle \text{ (ground atom, one photon)}$$

## Effective Hamiltonian in Subspace

In the basis  $\{|1, 0\rangle, |0, 1\rangle\}$ , the Hamiltonian reduces to:

$$H_{\text{eff}} = \begin{pmatrix} -\Delta & g \\ g & \omega_c \end{pmatrix} \quad (19)$$

## Time Evolution

The state evolves as:

$$|\psi(t)\rangle = c_{1,0}(t) |1, 0\rangle + c_{0,1}(t) |0, 1\rangle \quad (20)$$

Solving the Schrödinger equation gives **Rabi oscillations** between the atom and the cavity photon:

$$|c_{1,0}(t)|^2 = \cos^2(\Omega t), \quad |c_{0,1}(t)|^2 = \sin^2(\Omega t) \quad (21)$$

with the **generalized Rabi frequency**:

$$\Omega = \sqrt{g^2 + \left(\frac{\Delta - \omega_c}{2}\right)^2} \quad (22)$$

## Observable $\langle\sigma^+\sigma^-\rangle(t)$

This operator is the **atomic excitation number operator** (projector onto the excited state). Its expectation value is the probability of the atom being excited:

$$\langle\sigma^+\sigma^-\rangle(t) = |c_{1,0}(t)|^2 = \cos^2(\Omega t) \quad (23)$$

At resonance ( $\Delta = \omega_c$ ):

$$\langle\sigma^+\sigma^-\rangle(t) = \cos^2(gt) \quad (24)$$

## 5 Squeezed States and the Uncertainty Principle

To show that a coherent squeezed state  $|\alpha, \xi\rangle$  saturates the uncertainty principle, we calculate the variances of the quadrature operators  $\hat{x}$  and  $\hat{y}$  (often denoted  $X_1$  and  $X_2$ ):

$$\hat{x} = \frac{1}{2}(\hat{a} + \hat{a}^\dagger), \quad \hat{y} = \frac{1}{2i}(\hat{a} - \hat{a}^\dagger) \quad (25)$$

## Squeezing Operator Transformations

The state is defined as  $|\alpha, \xi\rangle = D(\alpha)S(\xi)|0\rangle$ . Under the squeezing operator  $S(\xi)$  with  $\xi = re^{i\theta}$ :

- $S^\dagger(\xi)\hat{a}S(\xi) = \hat{a} \cosh r - \hat{a}^\dagger e^{i\theta} \sinh r$
- $S^\dagger(\xi)\hat{a}^\dagger S(\xi) = \hat{a}^\dagger \cosh r - \hat{a} e^{-i\theta} \sinh r$

The displacement operator  $D(\alpha)$  shifts the mean values ( $\langle\hat{x}\rangle$  and  $\langle\hat{y}\rangle$ ) but **does not change the variances**. Therefore, we evaluate the variances for the squeezed vacuum state  $S(\xi)|0\rangle$ .

## Calculating the Variances

Assuming  $\theta = 0$  (real squeezing parameter), the transformed operators become:

$$\hat{x}' = S^\dagger \hat{x} S = \hat{x} e^{-r}, \quad \hat{y}' = S^\dagger \hat{y} S = \hat{y} e^r \quad (26)$$

For the  $\hat{x}$  quadrature:

$$(\Delta x)^2 = \langle 0 | (\hat{x} e^{-r})^2 | 0 \rangle = e^{-2r} \langle 0 | \hat{x}^2 | 0 \rangle \quad (27)$$

Since the variance of the vacuum state for these quadratures is  $1/4$ :

$$(\Delta x)^2 = \frac{1}{4} e^{-2r} \quad (28)$$

For the  $\hat{y}$  quadrature:

$$(\Delta y)^2 = \langle 0 | (\hat{y} e^r)^2 | 0 \rangle = e^{2r} \langle 0 | \hat{y}^2 | 0 \rangle = \frac{1}{4} e^{2r} \quad (29)$$

Thus the uncertainty product saturates the Heisenberg bound:

$$\Delta x \cdot \Delta y = \frac{1}{4} \quad (30)$$

## 6 The Lindblad Equation

The **Lindblad equation** (or GKLS equation) is the mathematical framework used to describe how a quantum state evolves when it is “open” to its environment.

### The Equation

$$\frac{d\rho}{dt} = -i[H, \rho] + \sum_n \left( L_n \rho L_n^\dagger - \frac{1}{2} \{L_n^\dagger L_n, \rho\} \right) \quad (31)$$

- $-i[H, \rho]$ : The **Unitary Part** — standard internal evolution.
- The summation: **The Dissipator** — describes jump processes caused by the environment.
- $L_n$ : **Lindblad Operators** — represent specific physical processes like photon emission, dephasing, or electron tunneling.
- $[A, B]$  and  $\{A, B\}$ : commutator and anti-commutator, respectively.

## Properties (Markovian Assumption)

The Lindblad equation is **Markovian**, leading to three vital properties of the density matrix:

1. **Trace Preservation:**  $\text{Tr}(\rho) = 1$  at all times.
2. **Hermiticity:**  $\rho = \rho^\dagger$ .
3. **Positivity:**  $\rho \geq 0$  (no negative probabilities).

## Example: Pure Dephasing

For a system undergoing pure dephasing with Lindblad operator  $L = \sqrt{\gamma} \sigma_z$ :

$$\dot{\rho} = -\Gamma_\phi \left( \sigma_z \rho \sigma_z - \frac{1}{2} \{ \sigma_z^2, \rho \} \right) \quad (32)$$

Since  $\sigma_z^2 = 1$ , this simplifies to:

$$\dot{\rho} = -\Gamma_\phi (\sigma_z \rho \sigma_z - \rho) \quad (33)$$

Using the matrix representation of  $\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ :

$$\sigma_z \rho \sigma_z - \rho = \begin{pmatrix} 0 & -2\rho_{01} \\ -2\rho_{10} & 0 \end{pmatrix} \quad (34)$$

This gives:

- $\dot{\rho}_{00} = 0$
- $\dot{\rho}_{11} = 0$
- $\dot{\rho}_{01} = -2\Gamma_\phi \rho_{01} \implies \rho_{01}(t) = \rho_{01}(0) e^{-2\Gamma_\phi t}$

The time evolution of the full density matrix is:

$$\rho(t) = \begin{pmatrix} |\alpha|^2 & \alpha\beta^* e^{-\gamma t} \\ \alpha^*\beta e^{-\gamma t} & |\beta|^2 \end{pmatrix} \quad (35)$$

with  $2\Gamma_\phi = \gamma$ .

## 7 Kraus Operators

**Kraus operators** are a set of operators  $\{K_i\}$  used to describe the most general evolution of a quantum system interacting with an environment. The **Kraus Representation Theorem** expresses the density matrix evolution as:

$$\rho(t) = \mathcal{E}(\rho) = \sum_i K_i \rho K_i^\dagger \quad (36)$$

To ensure trace preservation (total probability = 1), the operators must satisfy the **completeness relation**:

$$\sum_i K_i^\dagger K_i = \mathbf{1} \quad (37)$$



## Connection to the Lindblad Equation

For a small time step  $\delta t$ , the Kraus operators are related to the Lindblad operators  $L_i$  and Hamiltonian  $H$ :

- $K_0 = \mathbf{1} + \left(\frac{-i}{\hbar}H + W\right) \delta t$ , where  $W = -\frac{1}{2} \sum_{i \neq 0} L_i^\dagger L_i$
- $K_i = L_i \sqrt{\delta t}$

## Example: Pure Dephasing

For a system undergoing pure dephasing with rate  $\gamma$ , the Kraus operators for a small time step  $\delta t$  are:

- $K_0 = \sqrt{1 - \gamma \delta t} \mathbf{1} \approx \left(1 - \frac{\gamma}{2} \delta t\right) \mathbf{1}$  (the “no-jump” operator)
- $K_1 = \sqrt{\gamma \delta t} \sigma_z$  (the “jump” operator)

These satisfy  $K_0^\dagger K_0 + K_1^\dagger K_1 = \mathbf{1}$  to first order in  $\delta t$ .

## 8 Jordan–Wigner Transformation

The **Jordan–Wigner transformation** is a mathematical tool used to map a one-dimensional system of spins (like the Ising or Heisenberg models) into a system of non-interacting fermions.

### The Core Problem: Statistics

In quantum mechanics, spin operators at different sites commute:

$$[\sigma_i, \sigma_j] = 0 \quad (i \neq j) \quad (38)$$

However, fermionic operators must anti-commute:

$$\{c_i, c_j^\dagger\} = \delta_{ij} \quad (39)$$

### The Mapping

The transformation replaces spin operators with fermionic creation ( $c_j^\dagger$ ) and annihilation ( $c_j$ ) operators:

- **Spin to Fermion:**  $\sigma_j^+ = c_j^\dagger \exp\left(i\pi \sum_{l < j} c_l^\dagger c_l\right)$
- **Fermion to Spin:**  $c_j = \left(\prod_{l < j} \sigma_l^z\right) \sigma_j^-$

The exponential term (the “string”) ensures that when you swap two fermions, you pick up a minus sign, satisfying the Pauli exclusion principle and anti-commutation rules.

## Fermions vs. Bosons

| Feature    | Fermions                           | Bosons                       |
|------------|------------------------------------|------------------------------|
| Spin       | Half-integer ( $1/2, 3/2, \dots$ ) | Integer ( $0, 1, 2, \dots$ ) |
| Statistics | Fermi–Dirac                        | Bose–Einstein                |
| Exclusion? | Yes (Pauli Exclusion Principle)    | No                           |
| Role       | Matter (building blocks)           | Forces (interactions)        |

## Why It Matters

- **Simplification:** Turns a complex “many-body” spin problem into a “free fermion” problem.
- **Diagonalization:** Allows finding the exact spectrum  $\epsilon_k$ .
- **Open Systems:** In the context of the Lindblad equation, helps identify the correct  $L_i$  operators when the environment interacts with the “fermionic” quasiparticles of the system.